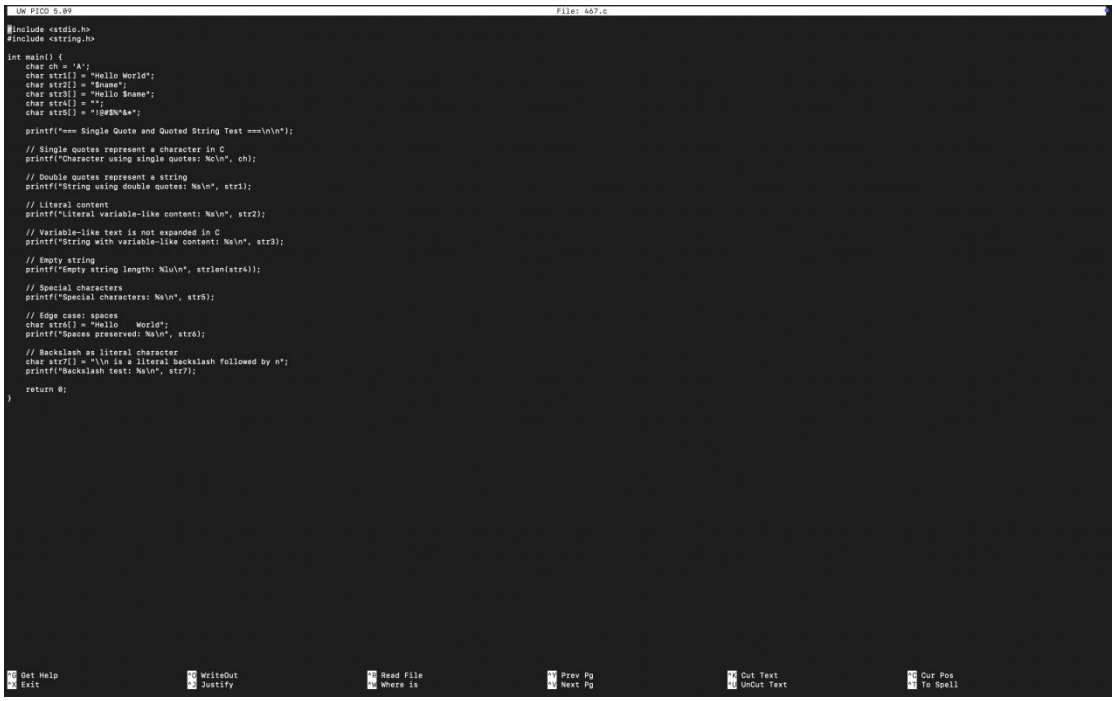
Name: B SREE NIKHIL REDDY ID number: 2520090058

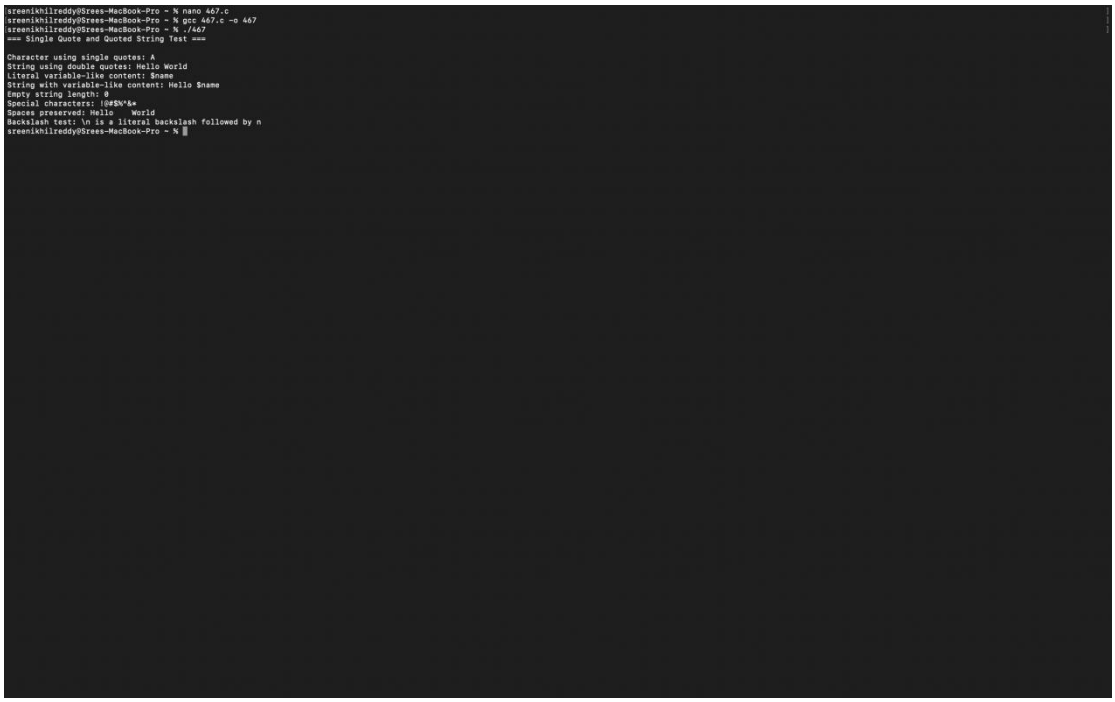
Week 10

Task 1: To Apply Single Quotes, Preserve Literal Content, Ignore Variable Expansion, Store Quoted Strings, Validate Parsing Results, Test Edge Cases.

CODE:

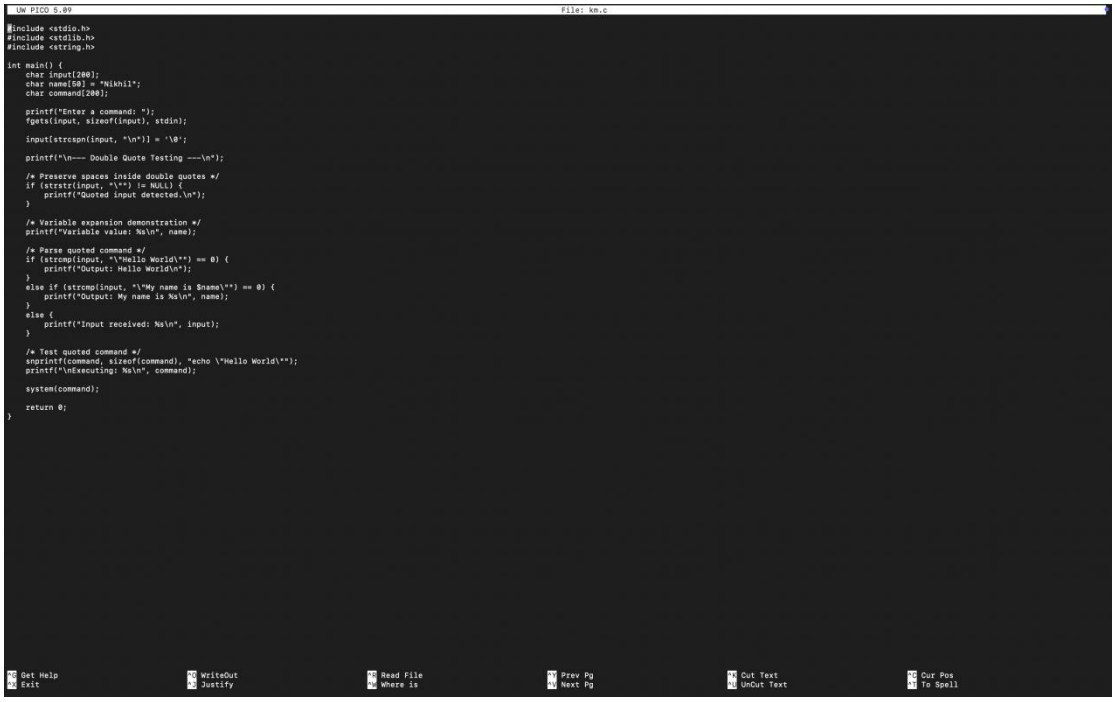


OUTPUT:

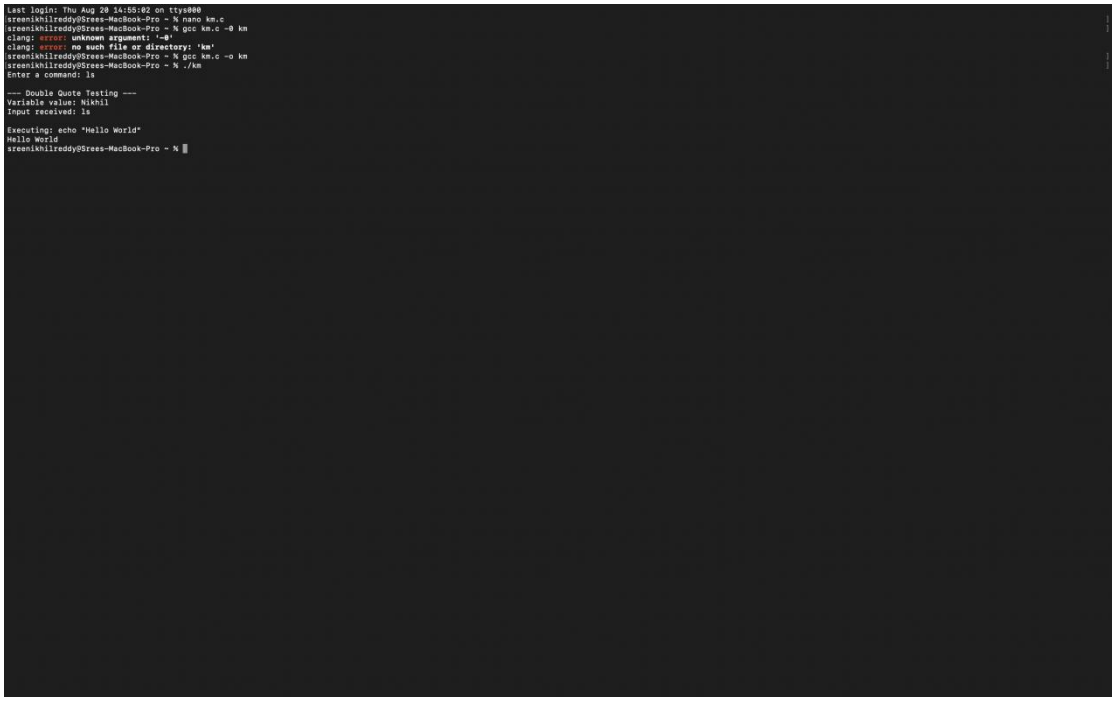


Task 2: To Apply Double Quotes, Preserve Spaces, Allow Variable Expansion, Parse Nested Tokens, Validate Outputs, Test Quoted Commands.

CODE:



OUTPUT:

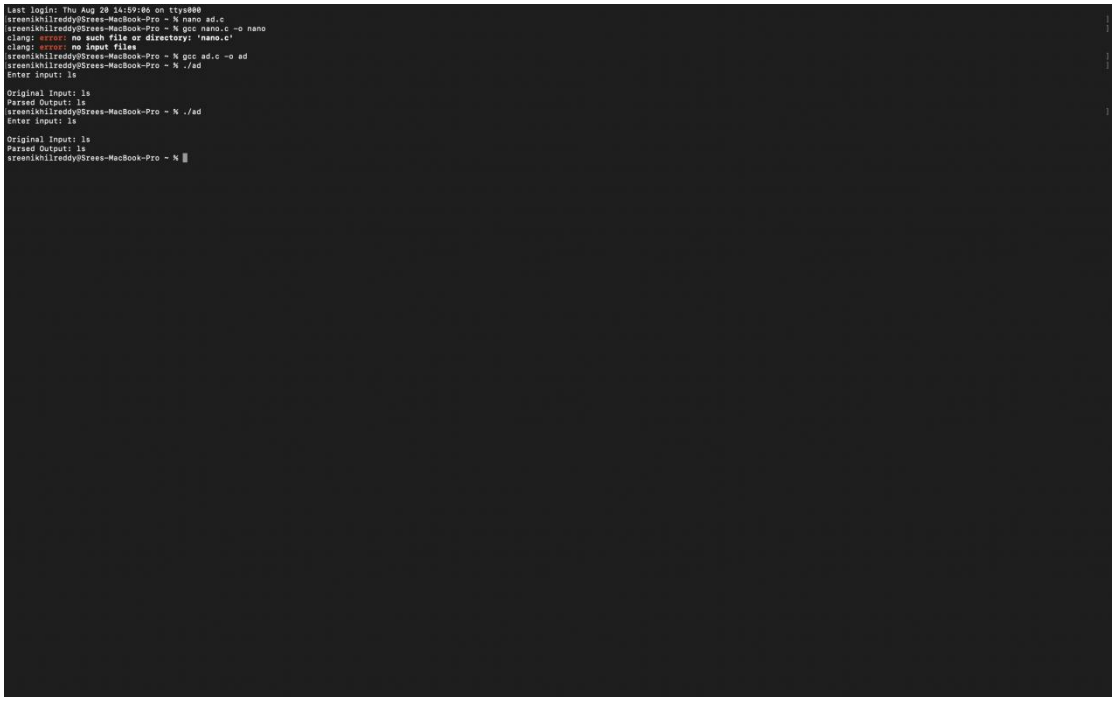


Task 1: To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs.

CODE:



OUTPUT:



Task 2: To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching

CODE:



```
  UW PICO 5.09                                   File: c.c

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid;
    int status;

    printf("Parent Process Started\n");
    printf("Parent PID: %d\n", getpid());

    /* Create child process */
    pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {
        /* Child process */
        printf("\nChild Process Created\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        /* Execute program with arguments */
        printf("Executing: ls -l\n\n");

        execlp("ls", "ls", "-l", NULL);

        /* Executes only if execlp fails */
        perror("Execution failed");
        exit(1);
    }
    else {
        /* Parent process */
        printf("\nParent waiting for child...\n");

        if (waitpid(pid, &status, 0) == -1) {
            perror("waitpid failed");
            return 1;
        }

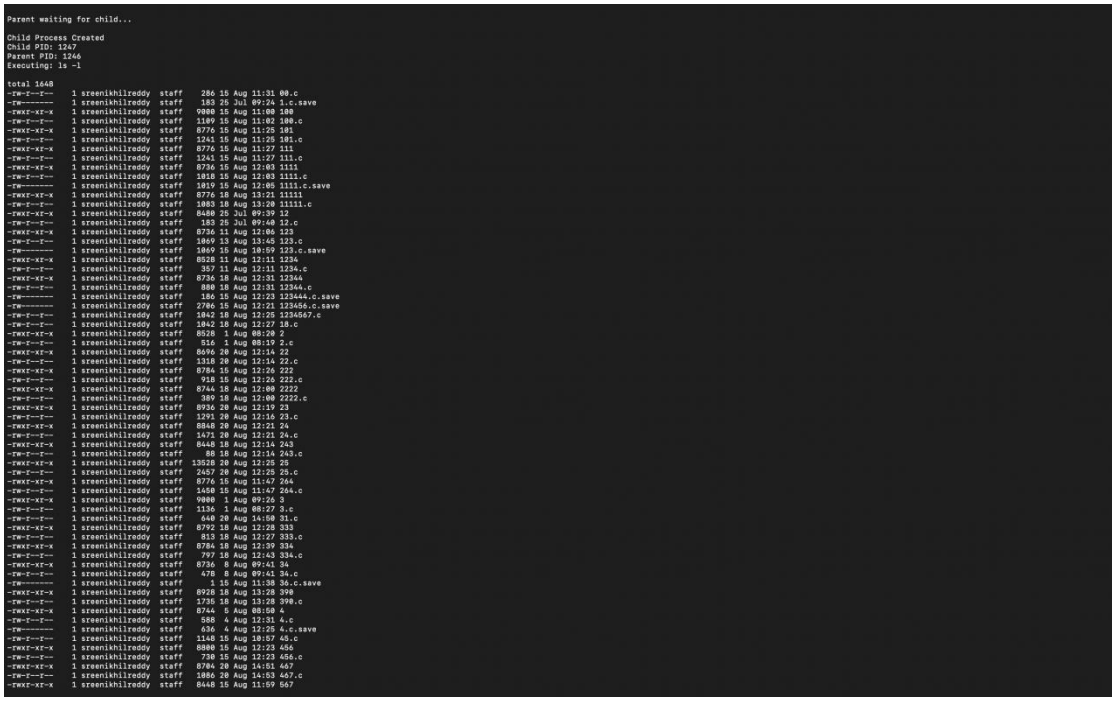
        if (WIFEXITED(status)) {
            printf("\nChild exited with status: %d\n",
                   WEXITSTATUS(status));
        }
        else {
            printf("\nChild terminated abnormally\n");
        }

        printf("Parent Process Finished\n");
    }

    return 0;
}

^G Get Help   ^O WriteOut   ^R Read File   ^Y Prev Pg   ^K Cut Text   ^C Cur Pos
^X Exit       ^J Justify    ^W Where is    ^V Next Pg   ^U UnCut Text ^T To Spell
```

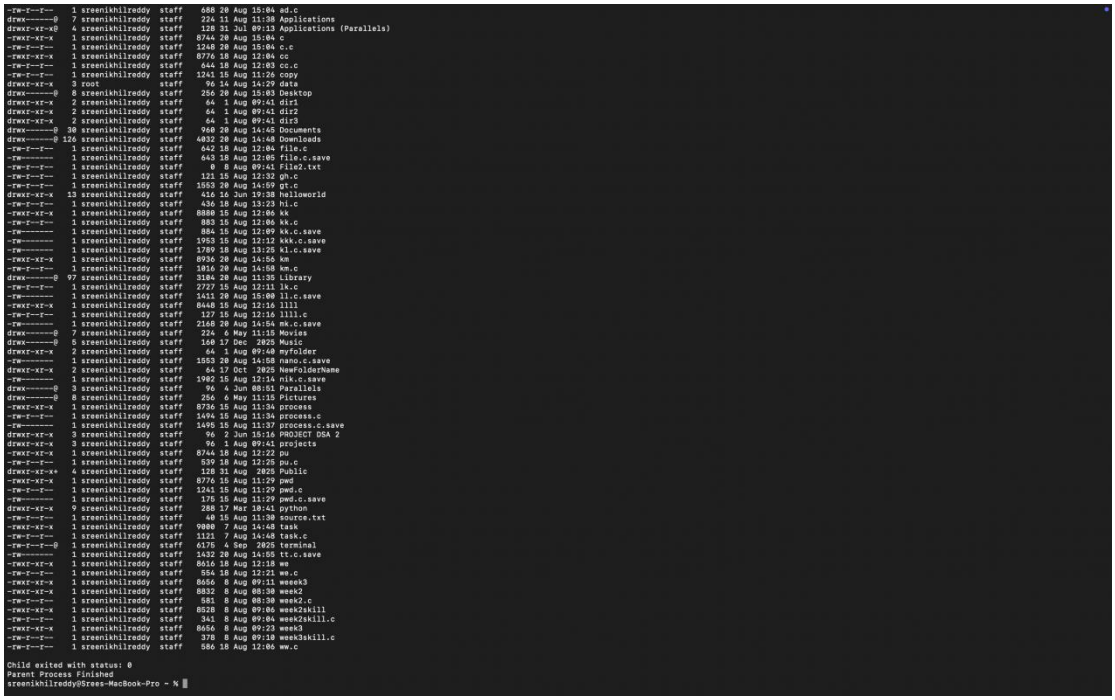
OUTPUT:



```
Parent waiting for child...

Child Process Created
Child PID: 1247
Parent PID: 1246
Executing: ls -l

total 1648
-rw-r--r--@   1 sreenikhilreddy  staff    286 15 Aug 11:31 00.c
-rw-------    1 sreenikhilreddy  staff    183 26 Jul 09:24 1.c.save
-rwxr-xr-x    1 sreenikhilreddy  staff   9080 15 Aug 11:00 100
-rw-r--r--@   1 sreenikhilreddy  staff   1189 15 Aug 11:02 100.c
-rwxr-xr-x    1 sreenikhilreddy  staff   8776 15 Aug 11:25 101
-rw-r--r--@   1 sreenikhilreddy  staff   1241 15 Aug 11:25 101.c
-rwxr-xr-x    1 sreenikhilreddy  staff   8776 15 Aug 11:27 111
-rw-r--r--@   1 sreenikhilreddy  staff   1241 15 Aug 11:27 111.c
-rwxr-xr-x    1 sreenikhilreddy  staff   8736 15 Aug 12:03 1111
-rw-r--r--@   1 sreenikhilreddy  staff   1018 15 Aug 12:03 1111.c
-rw-------    1 sreenikhilreddy  staff   1019 15 Aug 12:05 1111.c.save
-rwxr-xr-x    1 sreenikhilreddy  staff   8776 18 Aug 13:21 11111
-rw-r--r--@   1 sreenikhilreddy  staff   1083 18 Aug 13:20 11111.c
-rwxr-xr-x    1 sreenikhilreddy  staff   8480 25 Jul 09:39 12
-rw-r--r--    1 sreenikhilreddy  staff    183 25 Jul 09:40 12.c
-rwxr-xr-x    1 sreenikhilreddy  staff   8736 11 Aug 12:06 123
-rw-r--r--@   1 sreenikhilreddy  staff   1069 13 Aug 13:45 123.c
-rw-------    1 sreenikhilreddy  staff   1069 15 Aug 10:59 123.c.save
-rwxr-xr-x    1 sreenikhilreddy  staff   8928 11 Aug 12:11 1234
-rw-r--r--@   1 sreenikhilreddy  staff    357 11 Aug 12:11 1234.c
-rwxr-xr-x    1 sreenikhilreddy  staff   8736 18 Aug 12:31 12344
-rw-r--r--@   1 sreenikhilreddy  staff    880 18 Aug 12:31 12344.c
-rw-------    1 sreenikhilreddy  staff    186 15 Aug 12:23 123444.c.save
-rw-------    1 sreenikhilreddy  staff   2706 15 Aug 12:21 123456.c.save
-rw-r--r--@   1 sreenikhilreddy  staff   1042 18 Aug 12:25 1234567.c
-rw-r--r--@   1 sreenikhilreddy  staff   1042 18 Aug 12:27 18.c
-rwxr-xr-x    1 sreenikhilreddy  staff   8528  1 Aug 08:20 2
-rw-r--r--@   1 sreenikhilreddy  staff    516  1 Aug 08:19 2.c
-rwxr-xr-x    1 sreenikhilreddy  staff   8696 20 Aug 12:14 22
-rw-r--r--@   1 sreenikhilreddy  staff   1318 20 Aug 12:14 22.c
-rwxr-xr-x    1 sreenikhilreddy  staff   8784 15 Aug 12:26 222
-rw-r--r--@   1 sreenikhilreddy  staff    918 15 Aug 12:26 222.c
-rwxr-xr-x    1 sreenikhilreddy  staff   8744 18 Aug 12:00 2222
-rw-r--r--@   1 sreenikhilreddy  staff    389 18 Aug 12:00 2222.c
-rwxr-xr-x    1 sreenikhilreddy  staff   8936 20 Aug 12:19 23
-rw-r--r--@   1 sreenikhilreddy  staff   1291 20 Aug 12:16 23.c
-rwxr-xr-x    1 sreenikhilreddy  staff   8848 20 Aug 12:21 24
-rw-r--r--@   1 sreenikhilreddy  staff   1471 20 Aug 12:21 24.c
-rwxr-xr-x    1 sreenikhilreddy  staff   8448 18 Aug 12:14 243
-rw-r--r--@   1 sreenikhilreddy  staff     88 18 Aug 12:14 243.c
-rwxr-xr-x    1 sreenikhilreddy  staff  13528 20 Aug 12:25 25
-rw-r--r--@   1 sreenikhilreddy  staff   2457 20 Aug 12:25 25.c
-rwxr-xr-x    1 sreenikhilreddy  staff   8776 15 Aug 11:47 264
-rw-r--r--@   1 sreenikhilreddy  staff   1458 15 Aug 11:47 264.c
-rwxr-xr-x    1 sreenikhilreddy  staff   9080  1 Aug 09:26 3
-rw-r--r--@   1 sreenikhilreddy  staff   1136  1 Aug 08:27 3.c
-rw-r--r--@   1 sreenikhilreddy  staff    640 20 Aug 14:50 31.c
-rwxr-xr-x    1 sreenikhilreddy  staff   8792 18 Aug 12:28 333
-rw-r--r--@   1 sreenikhilreddy  staff    813 18 Aug 12:27 333.c
-rwxr-xr-x    1 sreenikhilreddy  staff   8784 18 Aug 12:39 334
-rw-r--r--@   1 sreenikhilreddy  staff    797 18 Aug 12:43 334.c
-rwxr-xr-x    1 sreenikhilreddy  staff   8736  8 Aug 09:41 34
-rw-r--r--@   1 sreenikhilreddy  staff    478  8 Aug 09:41 34.c
-rw-------    1 sreenikhilreddy  staff      1 15 Aug 11:38 36.c.save
-rwxr-xr-x    1 sreenikhilreddy  staff   8928 18 Aug 13:28 390
-rw-r--r--@   1 sreenikhilreddy  staff   1736 18 Aug 13:28 390.c
-rwxr-xr-x    1 sreenikhilreddy  staff   8744  5 Aug 08:50 4
-rw-r--r--@   1 sreenikhilreddy  staff    588  4 Aug 12:31 4.c
-rw-------    1 sreenikhilreddy  staff    636  4 Aug 12:25 4.c.save
-rw-r--r--@   1 sreenikhilreddy  staff   1148 15 Aug 10:57 45.c
-rwxr-xr-x    1 sreenikhilreddy  staff   8800 15 Aug 12:23 456
-rw-r--r--@   1 sreenikhilreddy  staff    730 15 Aug 12:23 456.c
-rwxr-xr-x    1 sreenikhilreddy  staff   8784 20 Aug 14:51 467
-rw-r--r--@   1 sreenikhilreddy  staff   1086 20 Aug 14:53 467.c
-rwxr-xr-x    1 sreenikhilreddy  staff   8448 16 Aug 11:59 567
```


-rw-r--r-- 1 sreenikhilreddy staff   688 20 Aug 15:04 ad.c
drwx------@ 7 sreenikhilreddy staff   224 11 Aug 11:38 Applications
drwxr-xr-x@ 4 sreenikhilreddy staff   128 31 Jul 09:13 Applications (Parallels)
-rwxr-xr-x   1 sreenikhilreddy staff 8744 20 Aug 15:04 c
-rw-r--r--   1 sreenikhilreddy staff 1248 20 Aug 15:04 c.c
-rwxr-xr-x   1 sreenikhilreddy staff 8776 18 Aug 12:04 cc
-rw-r--r--   1 sreenikhilreddy staff   644 18 Aug 12:03 cc.c
-rw-r--r--   1 sreenikhilreddy staff 1241 15 Aug 11:26 copy
drwxr-xr-x   3 root   staff   96 14 Aug 14:29 data
drwx------@ 8 sreenikhilreddy staff   256 20 Aug 15:03 Desktop
drwxr-xr-x   2 sreenikhilreddy staff   64 1 Aug 09:41 dir1
drwxr-xr-x   2 sreenikhilreddy staff   64 1 Aug 09:41 dir2
drwxr-xr-x   2 sreenikhilreddy staff   64 1 Aug 09:41 dir3
drwx------@ 30 sreenikhilreddy staff   960 20 Aug 14:45 Documents
drwx------@ 126 sreenikhilreddy staff 4032 20 Aug 14:48 Downloads
-rw-r--r--   1 sreenikhilreddy staff   642 18 Aug 12:04 file.c
-rw-------   1 sreenikhilreddy staff   643 18 Aug 12:05 file.c.save
-rw-r--r--   1 sreenikhilreddy staff   0 8 Aug 09:41 File2.txt
-rw-r--r--   1 sreenikhilreddy staff   121 15 Aug 12:52 gh.c
-rw-r--r--   1 sreenikhilreddy staff 1553 20 Aug 14:59 gt.c
drwxr-xr-x 13 sreenikhilreddy staff   416 16 Jun 19:38 helloworld
-rw-r--r--   1 sreenikhilreddy staff   436 18 Aug 13:23 hi.c
-rwxr-xr-x   1 sreenikhilreddy staff 8880 15 Aug 12:06 kk
-rw-r--r--   1 sreenikhilreddy staff   883 15 Aug 12:06 kk.c
-rw-------   1 sreenikhilreddy staff   884 15 Aug 12:09 kk.c.save
-rw-------   1 sreenikhilreddy staff 1953 15 Aug 12:12 kkk.c.save
-rw-------   1 sreenikhilreddy staff 1789 18 Aug 13:25 kl.c.save
-rwxr-xr-x   1 sreenikhilreddy staff 8936 20 Aug 14:56 km
-rw-r--r--   1 sreenikhilreddy staff 1016 20 Aug 14:58 km.c
drwx------@ 97 sreenikhilreddy staff 3104 20 Aug 11:35 Library
-rw-r--r--   1 sreenikhilreddy staff 2727 15 Aug 12:11 lk.c
-rw-------   1 sreenikhilreddy staff 1411 20 Aug 15:00 ll.c.save
-rwxr-xr-x   1 sreenikhilreddy staff 8448 15 Aug 12:16 llll
-rw-r--r--   1 sreenikhilreddy staff   127 15 Aug 12:16 llll.c
-rw-------   1 sreenikhilreddy staff 2168 20 Aug 14:54 mk.c.save
drwx------@ 7 sreenikhilreddy staff   224 6 May 11:15 Movies
drwx------@ 5 sreenikhilreddy staff   160 17 Dec 2025 Music
drwxr-xr-x   2 sreenikhilreddy staff   64 1 Aug 09:40 myfolder
-rw-------   1 sreenikhilreddy staff 1553 20 Aug 14:58 nano.c.save
drwxr-xr-x   2 sreenikhilreddy staff   64 17 Oct 2025 NewFolderName
-rw-------   1 sreenikhilreddy staff 1982 15 Aug 12:14 nik.c.save
drwx------@ 3 sreenikhilreddy staff   96 4 Jun 08:51 Parallels
drwx------@ 8 sreenikhilreddy staff   256 6 May 11:15 Pictures
-rwxr-xr-x   1 sreenikhilreddy staff 8736 15 Aug 11:34 process
-rw-r--r--   1 sreenikhilreddy staff 1494 15 Aug 11:34 process.c
-rw-------   1 sreenikhilreddy staff 1495 15 Aug 11:37 process.c.save
drwxr-xr-x   3 sreenikhilreddy staff   96 2 Jun 15:16 PROJECT DSA 2
drwxr-xr-x   3 sreenikhilreddy staff   96 1 Aug 09:41 projects
-rwxr-xr-x   1 sreenikhilreddy staff 8744 18 Aug 12:22 pu
-rw-r--r--   1 sreenikhilreddy staff   539 18 Aug 12:25 pu.c
drwxr-xr-x+ 4 sreenikhilreddy staff   128 31 Aug 2025 Public
-rwxr-xr-x   1 sreenikhilreddy staff 8776 15 Aug 11:29 pwd
-rw-r--r--   1 sreenikhilreddy staff 1241 15 Aug 11:29 pwd.c
-rw-------   1 sreenikhilreddy staff   175 15 Aug 11:29 pwd.c.save
drwxr-xr-x   9 sreenikhilreddy staff   288 17 Mar 10:41 python
-rw-r--r--   1 sreenikhilreddy staff   40 15 Aug 11:30 source.txt
-rwxr-xr-x   1 sreenikhilreddy staff 9000 7 Aug 14:48 task
-rw-r--r--   1 sreenikhilreddy staff 1121 7 Aug 14:48 task.c
-rw-r--r--@ 1 sreenikhilreddy staff 6175 4 Sep 2025 terminal
-rw-------   1 sreenikhilreddy staff 1432 20 Aug 14:55 tt.c.save
-rwxr-xr-x   1 sreenikhilreddy staff 8616 18 Aug 12:18 we
-rw-r--r--   1 sreenikhilreddy staff   554 18 Aug 12:21 we.c
-rwxr-xr-x   1 sreenikhilreddy staff 8656 8 Aug 09:11 week3
-rwxr-xr-x   1 sreenikhilreddy staff 8832 8 Aug 08:30 week2
-rw-r--r--   1 sreenikhilreddy staff   581 8 Aug 08:30 week2.c
-rwxr-xr-x   1 sreenikhilreddy staff 8528 8 Aug 09:04 week2skill
-rw-r--r--   1 sreenikhilreddy staff   341 8 Aug 09:04 week2skill.c
-rwxr-xr-x   1 sreenikhilreddy staff 8656 8 Aug 09:23 week3
-rw-r--r--   1 sreenikhilreddy staff   378 8 Aug 09:18 week3skill.c
-rw-r--r--   1 sreenikhilreddy staff   586 18 Aug 12:06 ww.c
Child exited with status: 0
Parent Process Finished
sreenikhilreddy@Srees-MacBook-Pro ~ %